Maulana Azad National Institute of Technology Bhopal
Department of Physics
End Term Examinations (B. Tech. II[nd] Sem.), May 2025

Subject: Physics **Subject code: PHY24102** **Duration: 2 hrs.** **Max. Marks: 40**

Note: Answer all questions. **Section: F, G, H, I & J**

Q.No	Questions	Marks	COs
1a	Explain the role of time-varying magnetic field in Betatron.	1	CO1
1b	What is zero-point energy of a simple quantum harmonic oscillator? Explain its physical significance.	2	
1d	Why is a photodiode operated in reverse bias?	1	
2a	Derive the expression for the optical path difference in the interference pattern produced by light reflecting from a thin film of uniform thickness with suitable diagram, and explain the conditions for constructive and destructive interference.	4	CO2
2b	Determine the number of visible diffraction orders for light with a wavelength of 5000 Å incident on a grating having 2620 lines per inch.	2	CO3
3a	An electron is moving in a one-dimensional box of width $a = 1$ Å. If the wave function of the first excited state $\psi(x) = \sqrt{\frac{2}{a}}\, Sin\left(\frac{2\pi x}{a}\right)$, then determine the probability of finding the electron between $x = 0$ to $x = 0.5$ Å in that quantum state.	4	CO3
3b	A 1.0 keV electron has its position measured with an uncertainty of 1 Å. What is the corresponding percentage uncertainty in its momentum?	2	CO4
4a	Derive an expression for the carrier concentration of electrons in the conduction band of an intrinsic semiconductors.	4	CO4
4b	For a *p-n-p* transistor in CE mode, $\beta = 100$. If $I_{CBO} = 15$ μA, calculate the collector current for an emitter current of 2 mA.	2	CO5
4c	The hall coefficient of a specimen of a doped silicon is found to be 3.66 x 10^{-4} m^3/C. The resistivity of the specimen is 8.93 x 10^{-3} Ω m. Find the mobility and density of the charge carriers.	3	CO5
5a	Describe the principle, construction, and working of a Cyclotron.	4	CO2
5b	In a Bainbridge mass spectrograph, the electric field used is 7×10^4 V/m and the magnetic field common to both places is 0.65 Wb/m^2. If the ion source consists of singly ionized chlorine isotopes of atomic masses 35 u and 37 u, calculate linear separation of lines formed on photographic plate (Given, 1u = 1.66×10^{-27} kg, 1 Wb/m^2 = 1 T).	3	CO5
6a	Derive the relationship between the Einstein A coefficient (A_{21}) for spontaneous emission and the Einstein B coefficient (B_{21}) for stimulated emission, considering the transition between two energy levels in an atom.	3	CO3
6b	Draw the energy level diagram of a He-Ne laser and briefly explain the principle of lasing action.	3	CO2
6c	The refractive indices for core and cladding for a step index fibre are 1.5 and 1.48, respectively. Calculate the values of (i) acceptance angle (θ_A) and (ii) numerical aperture.	2	CO4

Useful Information: $e = 1.6 \times 10^{-19}$ C

$k_B = 1.38 \times 10\text{-}23$ J/K

$m_e = 9.11 \times 10^{-31}$ kg

$h = 6.626 \times 10^{-34}$ J s

End